

YOUSSEF HASSAN

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Summary

Backend and embedded systems engineer with strong foundations in software engineering and Artificial Intelligence. Skilled in Python, C++, and C, with hands-on experience in building scalable and real-time systems. Experienced in problem-solving through competitive programming (ECPC), with a focus on efficient and reliable solutions.

Education

2022 – 2026

Bachelor's in Computer Science and Artificial Intelligence

Cairo University, Giza, Egypt | 2022 – 2026

Experience

Apr 2024 – Mar 2026

Prompt Engineer at Scale-AI Company

- Designed and optimized AI prompts to improve model performance and output quality
- Collaborated with teams to refine system responses and enhance accuracy
- Worked under pressure to meet performance and quality targets

Competitions & Certifications

2023. 2024

Egyptian Collegiate Programming Contest (ECPC) Qualifications

- Solved algorithmic problems under time constraints and strengthened problem-solving.

Projects

Movie Tracker SPA (Laravel, MVC, MySQL, PHPUnit, OMDb API)

- Developed a full-stack web application for tracking movies and ratings using the Laravel MVC framework. Built robust models with Eloquent ORM, dynamic views using Blade templating, and integrated the OMDb API for real-time data retrieval. Ensured code reliability through automated Feature and Unit testing with PHPUnit.

E-Commerce System API (.NET, ASP.NET Core, EF Core, SignalR, xUnit)

- Developed an enterprise-grade RESTful API following Clean Architecture principles, implementing secure Role-Based Access Control (RBAC), real-time order notifications using SignalR, and robust data access via Repository and Unit of Work patterns, while ensuring system reliability through comprehensive unit testing.

Inventory Management System (.NET, EF Core)

- Developed a scalable backend system for products, categories, and suppliers, implementing CRUD operations with filtering, sorting, and pagination, applying the Repository Pattern and Unit of Work, and designing a relational database with proper entity relationships.

School Management System (.NET, EF Core)

- Developed a backend system to manage students, courses, and supplies using Entity Framework Core with CRUD operations, applying data validation and business rules, and designing a modular architecture with Controllers, Models, and DbContext, and implemented filtering logic for data retrieval.

Door Locker Security System

- Designed and implemented a smart door control system using ATmega32, developing low-level drivers (UART, I2C, EEPROM, Motor Control), implementing secure password validation, and integrating PIR sensor and buzzer with real-time system control.

Distance Measuring System

- Developed a system using an ultrasonic sensor (HC-SR04) for distance measurement, implementing accurate timing and signal processing logic.

Fan Speed Controller with Temperature

- Built an embedded system to control fan speed based on temperature readings by interfacing the LM35 sensor and implementing PWM control, optimizing system performance for real-time responsiveness.

Stopwatch System

- Designed a digital stopwatch using microcontroller timers, implementing precise timing through interrupts and timer modules.

Courses

Aug 2024

Full Embedded Systems Diploma under the supervision of Engineer Mohamed Tarek

- Developed embedded projects using AVR (ATmega32), implementing low-level drivers and real-time applications, with hands-on experience in RTOS and embedded systems design.

Apr 2026

Backend Development Diploma (.NET)

- Built backend systems using ASP.NET Core and Entity Framework Core, implementing RESTful APIs and CRUD operations, applying the Repository Pattern and Unit of Work, and working with SQL databases, validation, filtering, caching, authentication, and SignalR.

Technical Skills

Programming Languages	C/C++, Python	C#, PHP
Backend and APIs	- ASP.NET Core - REST APIs - Authentication & Authorization	- Entity Framework Core - Web Development (HTML, CSS, JavaScript) - JWT
Databases	SQL Server (SSMS)	MySQL
Embedded Systems	- Embedded C - Interfacing	- AVR (ATmega32) - Communication Protocols (UART, SPI, I2C)
Core Concepts	- Data Structures and Algorithms - SOLID Principles - Problem Solving	- Object Oriented Programming (OOP) - Software Engineering
Tools	Visual Studio / Visual Studio Code	Git/GitHub
